

Graphing Quadratics using Roots

Example Problems

EXAMPLE 1: Draw a graph of the quadratic equation $y = (x + 2)^2$, and mark all important features.

SOLUTION: First let's find the roots of the graph—when the graph crosses the x axis—by making $y = 0$,

$$0 = (x + 2)^2,$$

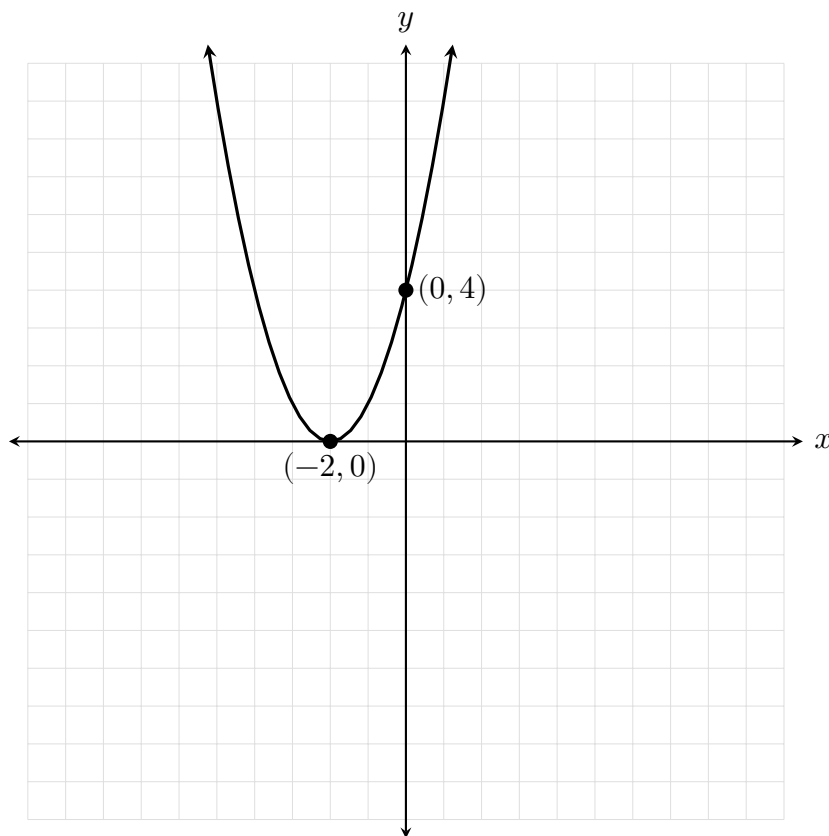
this will be true when $x = -2$. This means our graph will cross the x axis only once, and that for our quadratic this must be the turning point. Then we want to find the y -intercept, this will occur when $x = 0$,

$$y = (0 + 2)^2,$$

$$y = 2^2,$$

$$y = 4.$$

With this information we can plot the key points and sketch the graph,



EXAMPLE 2: Draw a graph of the quadratic equation $y = x^2 + 2x - 8$ and label all important features.

SOLUTION: First we must determine where the roots of the graph will be, and to do this we will factorise the quadratic expression. This gives

$$y = (x - 2)(x + 4),$$

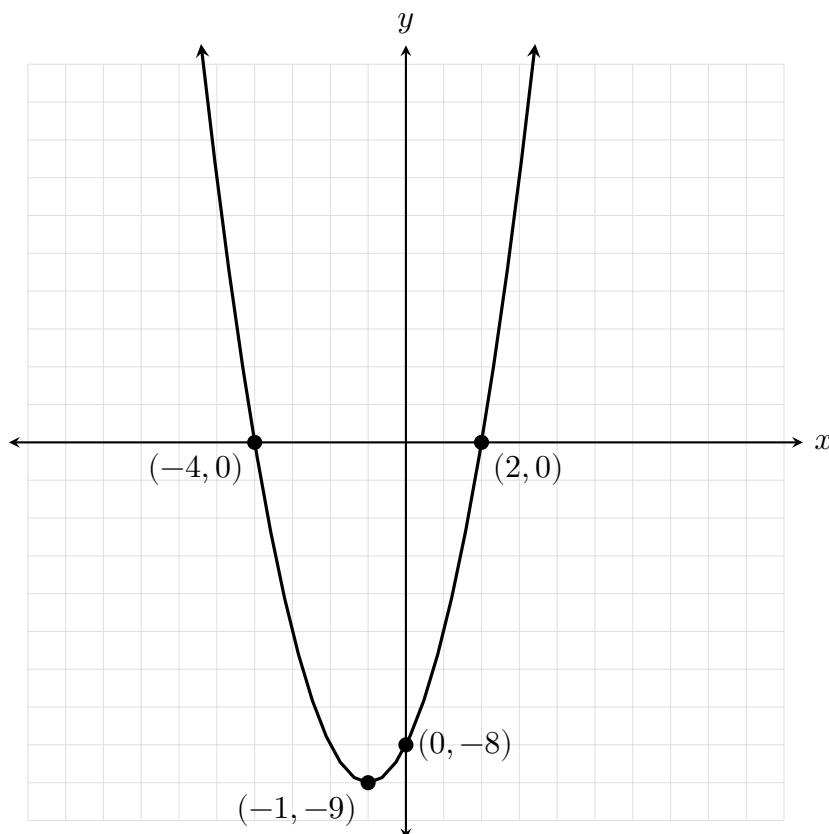
which tells us that the roots will occur at $x = 2$ and $x = -4$. Further we can find the y -intercept by substituting $x = 0$ to get

$$\begin{aligned}y &= 0^2 + 2 \times 0 - 8 \\y &= -8.\end{aligned}$$

Finally we will find the turning point. The turning point of a parabola always lies on its axis of symmetry. The axis of symmetry can be calculated either by finding the halfway point between the roots -4 and 2 —which would be -1 —or by using the formula $\frac{-b}{2}$ which is also -1 . We can then substitute the axis of symmetry as x to find the y coordinate of the turning point.

$$\begin{aligned}y &= (-1)^2 + 2 \times -1 - 8, \\y &= 1 - 2 - 8, \\y &= -9.\end{aligned}$$

Then we can plot these points and sketch the graph.



EXAMPLE 3: Draw a graph of the quadratic equation $y = -x^2 + x + 6$ and label all important features.

SOLUTION: First we might notice that the coefficient of x^2 is -1 , which means our parabola will be concave down—the arms will point downwards. To factorise the quadratic expression we will factor -1 out of the trinomial expression then factorise

it like any other quadratic,

$$y = -(x^2 - x - 6),$$
$$y = -(x - 3)(x + 2).$$

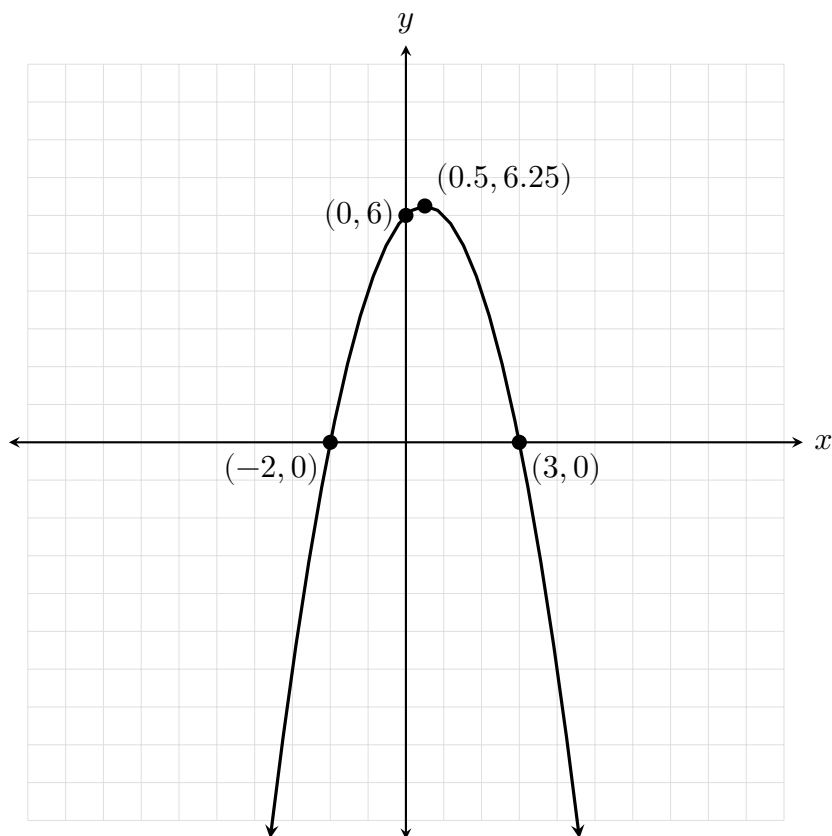
From this we can determine that the roots will be $x = -2$ and $x = 3$. We can find the y -intercept by substituting $x = 0$,

$$y = -0^2 + 0 + 6,$$
$$y = 6.$$

Finally, the x coordinate of the axis of symmetry will be 0.5, so we can find the y coordinate of the turning point to be

$$y = -0.5^2 + 0.5 + 6,$$
$$y = 6.25.$$

Then we can plot the key points and sketch the graph.



Question Bank

1. For each of the following quadratic equations find their roots, y -intercept, and turning point. Then graph the equation and label all the important points.

a) $y = x^2$

b) $y = x^2 + 3x + 2$

c) $y = x^2 + 2x + 1$

d) $y = x^2 - 4x + 3$

e) $y = x^2 - 6x + 9$

f) $y = x^2 + 11x + 28$

g) $y = x^2 - 14x + 45$

h) $y = x^2 - 3x - 4$

i) $y = x^2 + 2x - 24$

j) $y = x^2 - x - 56$

k) $y = -x^2 - 9x - 20$

l) $y = -x^2 - 2x - 1$

m) $y = -x^2 + 11x - 24$

n) $y = -x^2 + 8x - 16$

o) $y = -x^2 + 2x + 35$

p) $y = -x^2 - 3x + 10$